

Some Important Theorems in Number Theory

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Fermat's Little Theorem

Problem 1. Show that if n has $p - 1$ digits all equal to 1, where p is a prime not equal to 2, 3, or 5, then n is divisible by p .

Problem 2. Prove that for any prime $p > 17$, the number $p^{32} - 1$ is divisible by 16320.

Problem 3. Let p be an odd prime number. Show that if the equation $x^2 \equiv a \pmod{p}$ has a solution, then $a^{(p-1)/2} \equiv 1 \pmod{p}$. Conclude that there are infinitely many primes of the form $4m + 1$.

Problem 4. Prove that the equation $x^2 = y^3 + 7$ has no integer solutions.

Problem 5. Let $n > 1$ be a positive integer. Prove that the equation $(x + 1)^n - x^n = ny$ has no positive integer solutions.

Problem 6. Prove that the sequence $2^n - 3$, $n \geq 1$, contains an infinite subsequence whose terms are pairwise relatively prime.

Problem 7. Let (x_n) be a sequence of positive integers satisfying the recurrence relation

$$x_{n+1} = 5x_n - 6x_{n-1}.$$

Prove that infinitely many terms of the sequence are composite.

Willson Formula

Problem 8. For each positive integer n , find the greatest common divisor of $n! + 1$ and $(n + 1)!$.

Problem 9. Prove that there are no positive integers n such that the set

$$\{n, n + 1, n + 2, n + 3, n + 4, n + 5\}$$

can be partitioned into two sets with the product of the elements of one set equal to the product of the elements of the other set.

Problem 10. Let p be an odd prime. Show that if the equation $x^2 \equiv a \pmod{p}$ has no solution, then

$$a^{\frac{p-1}{2}} \equiv -1 \pmod{p}.$$

Problem 11. Let p be an odd prime number. Show that the equation $x^2 \equiv -1 \pmod{p}$ has a solution if and only if $p \equiv 1 \pmod{4}$.

Problem 12. Let p be a prime number and n an integer with $1 \leq n < p$. Prove that

$$(p-n)!(n-1)! \equiv (-1)^n \pmod{p}.$$

Problem 13. Let p be an odd prime and $a_1, a_2, \dots, a_p$ be an arithmetic progression whose common difference is not divisible by p . Prove that there exists an index i such that

$$a_1 a_2 \cdots a_p + a_i$$

is divisible by p^2 .

Euler's Theorem

Problem 14. Prove that for any positive integer n ,

$$\sum_{k|n} \varphi(k) = n,$$

where the sum is over all positive integers k dividing n .

Problem 15. Prove that for any positive integer n other than 2 or 6, $\varphi(n)$ is even.

Problem 16. Prove that there are infinitely many positive integers n such that

$$\varphi(n)^2 + n^2$$

is a perfect square.

Problem 17. Prove that there are infinitely many even positive integers m for which the equation $\varphi(n) = m$ has no solutions.

Problem 18. Prove that for every positive integer s , there exists a positive integer n divisible by s and with the sum of its digits equal to s .

Problem 19. Prove that the equation $2^x + 3 = z^3$ has no positive integer solutions.

Problem 20. Prove that for every positive integer n , the following identity holds:

$$\frac{\varphi(1)}{1} + \frac{\varphi(2)}{2} + \frac{\varphi(3)}{3} + \cdots + \frac{\varphi(n)}{n} = \frac{n(n+1)}{2}.$$

Problem 21. Given nonzero integers a and d , show that the arithmetic sequence

$$a, a + d, a + 2d, \dots$$

contains infinitely many terms that have the same prime factors.

Chinese Remainder Theorem

Problem 22. Prove that for every n , there exist n consecutive integers each of which is divisible by two different primes.

Problem 23. Let $P(x)$ be a polynomial with integer coefficients. For any positive integer m , let $N(m)$ denote the number of solutions to the equation $P(x) \equiv 0 \pmod{m}$. Show that if m_1 and m_2 are coprime integers, then

$$N(m_1 m_2) = N(m_1) N(m_2).$$

Problem 24. Alice and Bob play a game in which they take turns removing stones from a heap that initially has n stones. The number of stones removed at each turn must be one less than a prime number. The winner is the player who takes the last stone. Alice plays first. Prove that there are infinitely many n such that Bob has a winning strategy.

Problem 25. Show that there exists an increasing sequence $(a_n)_{n \geq 1}$ of positive integers such that for any $k \geq 0$, the sequence $k + a_n$, $n \geq 1$, contains only finitely many primes.

Problem 26. Is there a sequence of positive integers in which every positive integer occurs exactly once and for every $k = 1, 2, 3, \dots$ the sum of the first k terms is divisible by k ?

Problem 27. Prove that there exists a positive integer k such that $k \cdot 2^n + 1$ is composite for every positive integer n .

Problem 28. Let a and b be two positive integers such that for any positive integer n , $a^n + n$ divides $b^n + n$. Prove that $a = b$.

Problem 29. A lattice point $(x, y) \in \mathbb{Z}^2$ is said to lie in the interior of a triangle if it is strictly inside the triangle and not on its boundary. Find, with proof, the number of lattice points in the interior of the triangle with vertices $(0, 0)$, $(n, 0)$, $(0, n)$, where n is a positive integer.